



NORTH SYDNEY GIRLS HIGH SCHOOL

Preliminary Mathematics

Assessment Task 2

Term 2, 2017

Name: _____ **Mathematics Class:** _____

Time Allowed: 55 minutes + 2 minutes reading time

Total Marks:

Section I (5 marks) - **Pages 2 - 3**

Attempt Questions 1- 5

Answer on Multiple Choice Answer Sheet

Section II (43 marks) - **Pages 4 - 7**

Attempt Questions 6 - 9

Show all necessary working

	MC		Q6		Q7		Q8		Q9		Total
Functions	4, 5	/2	a-d	/10	a	/1					13
Trigonometry	2, 3	/2			b, c	/10	a, b	/4	a, b	/4	20
Plane Geometry	1	/1					c	/6	c	/6	13
		/5		/10		/11		/10		/10	/46

Section I

5 Marks

Attempt Questions 1 – 5

Allow about 7 minutes.

Use the multiple choice answer sheet for questions 1 – 5.

1. A polygon has an interior angle sum of 3960° . How many sides does it have?

- (A) 11
- (B) 20
- (C) 22
- (D) 24

2. Simplify $\frac{\cos(90 - \theta)}{\sin(90 - \theta)}$.

- (A) 1
- (B) $\cot \theta$
- (C) $\tan \theta$
- (D) $-\tan \theta$

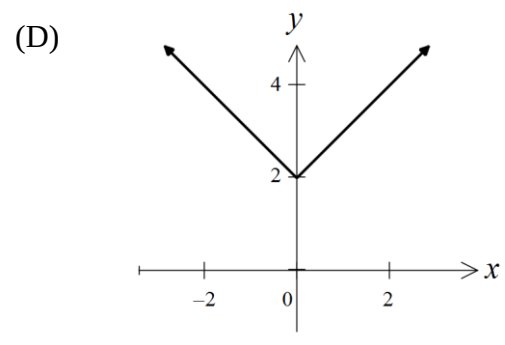
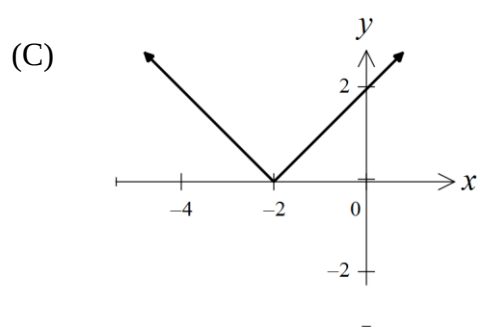
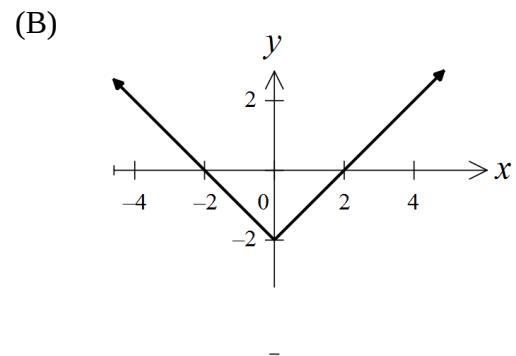
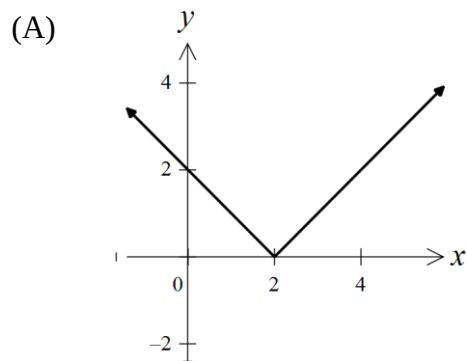
3. $\frac{2 - \tan 60^\circ}{\sec^2 45^\circ}$ in exact form is

- (A) $2\sqrt{3}$
- (B) $1 - \frac{\sqrt{3}}{2}$
- (C) $-\frac{1}{\sqrt{3}}$
- (D) $\frac{2 - \sqrt{3}}{\sqrt{2}}$

4. What is the domain and range of the function $f(x) = \sqrt{1-x^2}$?

- (A) Domain: $0 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
- (B) Domain: $-1 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
- (C) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq 1$
- (D) Domain: $0 \leq x \leq 1$, Range: $0 \leq y \leq 1$

5. Which graph best represents $y = |x| - 2$?



End of Section I

Section II **43 Marks**

Attempt Questions 6 – 9

Allow about 48 minutes.

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In questions 6 – 9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks)

Use a separate writing booklet

- (a) Is the function $f(x) = 8x^3 - 7x$ even, odd or neither? 2

Show working to support your answer.

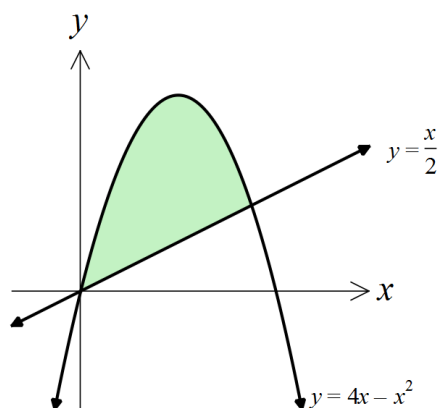
- (b) Find the centre and radius of the circle with equation $x(x-2) + y(y+10) = 23$. 2

- (c) On separate diagrams, sketch each of the following, showing all essential features.

(i) $y = \sqrt{x+7}$ 2

(ii) $y = 1 - 2^{-x}$ 2

- (d) The diagram shows the graphs of the functions $y = 4x - x^2$ and $y = \frac{1}{2}x$.



Write down a pair of inequalities that specify the shaded region. 2

End of Question 6

Question 7 (11 marks)

Use a separate writing booklet

- (a) The function $y = f(x)$ is defined as follows:

$$f(x) = \begin{cases} x-1 & \text{for } x \leq -2 \\ -1 & \text{for } -2 < x \leq 1 \\ x+1 & \text{for } x > 1 \end{cases}$$

Evaluate $f(-2) - f(17) + f(1)$.**1**

- (b) Solve each equation for $0^\circ \leq \theta \leq 360^\circ$, correct to the nearest degree where necessary.

(i) $\cos^2 \theta = \frac{3}{4}$ **2**

(ii) $\tan 2\theta = -5$ **2**

- (c) A ship sails 50km from port A to port B on a bearing of 063° T and then sails 130km from port B to port C on a bearing of 296° T.

- (i) Draw a diagram to represent this information. **1**

- (ii) Explain why $\angle ABC = 53^\circ$ **1**

- (iii) Find, correct to the nearest km, the distance from port A to port C. **2**

- (iv) Find the bearing of port A from port C. Answer to the nearest degree. **2**

End of Question 7

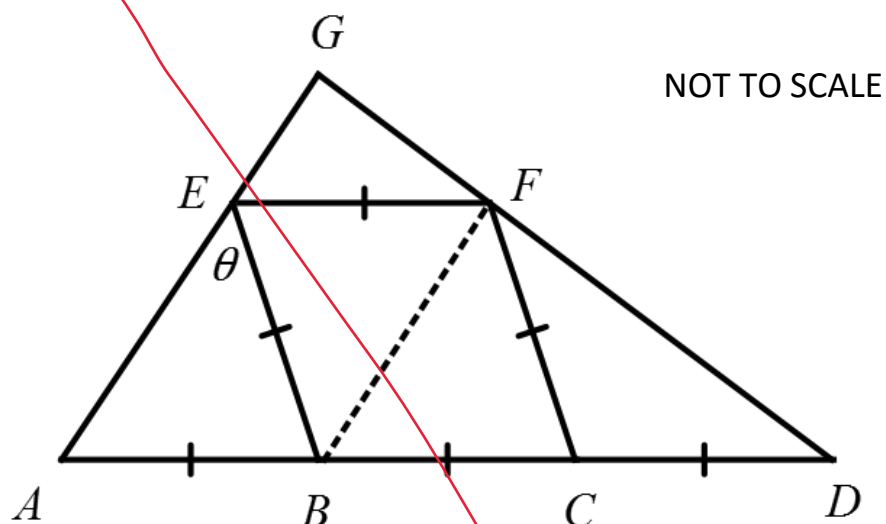
Question 8 (10 marks)

Use a separate writing booklet

- (a) Draw a neat sketch of $y = \operatorname{cosec} x$ for $0^\circ \leq x \leq 360^\circ$. 2

- (b) Prove that $\frac{\sin \beta}{1 + \cos \beta} = \frac{1 - \cos \beta}{\sin \beta}$. 2

- (c) $ABCD$ is a straight line such that $AB = BC = CD$.
 $EFCB$ is a rhombus and $\angle AEB = \theta$.



- (i) Show that $\angle EBC = 2\theta$. 2
- (ii) The diagonal BF is drawn in the rhombus $EFCB$.
Show that $BF \parallel AE$. 2
- (iii) Hence or otherwise, find the size of $\angle AGD$. 2
Show all working.

End of Question 8

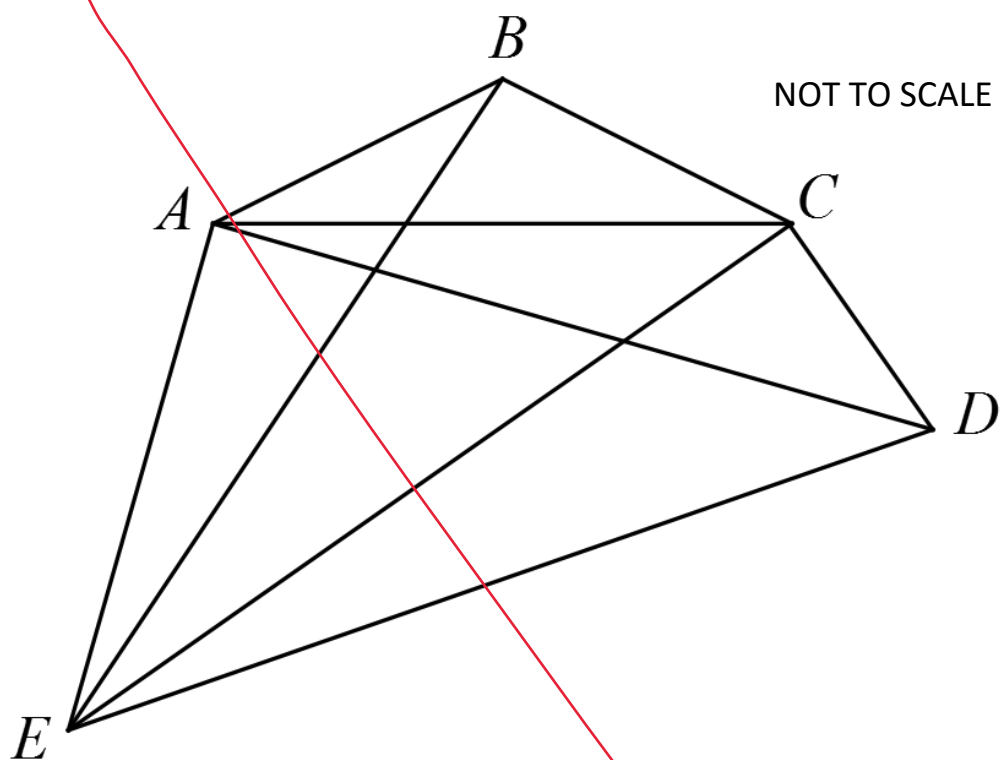
Question 9 (10 marks)

Use a separate writing booklet

(a) If β is a reflex angle such that $\tan \beta = \frac{2}{3}$, find the exact value of $\cos \beta$. **1**

(b) Solve $6\sin^2 x + 7\cos x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$. **3**
Give answers to the nearest degree.

- (c) In pentagon $ABCDE$, $AB = BC = CD$, $\angle AEC = \angle ACE$ and $\angle BAE + \angle BAC = \angle BCD$.



- (i) Explain why $\angle BAC = \angle BCA$ **1**
(ii) Show that $\angle BAE = \angle ACD$. **2**
(iii) Hence, prove that $\triangle EAB \equiv \triangle ACD$ **3**

End of Examination



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Plane Geometry	1	/1					c	/6	c	/6	13
		/5		/10		/11		/10		/10	/46

Section I

5 Marks

Attempt Questions 1 – 5

Allow about 7 minutes.

Use the multiple choice answer sheet for questions 1 – 5.

1. A polygon has an interior angle sum of 3960° . How many sides does it have?

- (A) 11
- (B) 20
- (C) 22
- (D) 24

$$\begin{aligned}(n-2)180 &= 3960 \\ n-2 &= 22 \\ n &= 24 \\ \text{Answer: D}\end{aligned}$$

2. Simplify $\frac{\cos(90-\theta)}{\sin(90-\theta)}$.

- (A) 1
- (B) $\cot \theta$
- (C) $\tan \theta$
- (D) $-\tan \theta$

$$\begin{aligned}&= \cot(90-\theta) \\ &= \tan \theta \\ \text{Answer: C}\end{aligned}$$

3. $\frac{2-\tan 60^\circ}{\sec^2 45^\circ}$ in exact form is

- (A) $2\sqrt{3}$
- (B) $1-\frac{\sqrt{3}}{2}$
- (C) $-\frac{1}{\sqrt{3}}$
- (D) $\frac{2-\sqrt{3}}{\sqrt{2}}$

$$\begin{aligned}&= \frac{2-\sqrt{3}}{(\sqrt{2})^2} \\ &= \frac{2-\sqrt{3}}{2} \\ \text{Answer: B}\end{aligned}$$

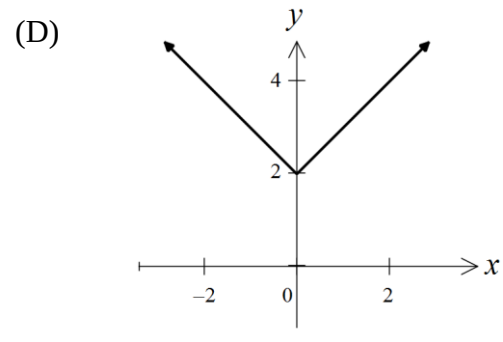
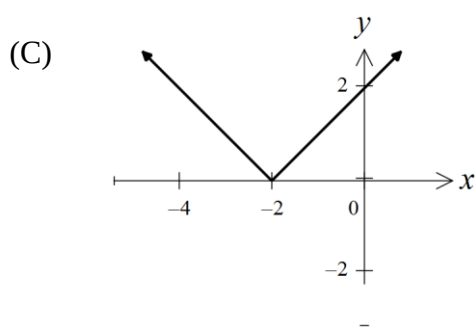
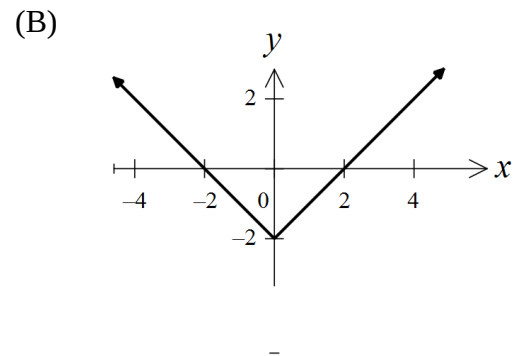
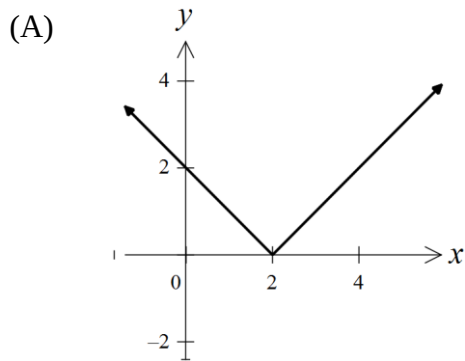
4. What is the domain and range of the function $f(x) = \sqrt{1-x^2}$?

- (A) Domain: $0 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
- (B) Domain: $-1 \leq x \leq 1$, Range: $-1 \leq y \leq 1$
- (C) Domain: $-1 \leq x \leq 1$, Range: $0 \leq y \leq 1$
- (D) Domain: $0 \leq x \leq 1$, Range: $0 \leq y \leq 1$

Answer: C

5. Which graph best represents $y = |x| - 2$?

Answer: B



End of Section I

Section II 43 Marks

Attempt Questions 6 – 9

Allow about 48 minutes.

Answer each question in a SEPARATE writing booklet. Extra writing paper is available.

In questions 6 – 9, you responses should include relevant mathematical reasoning and/or calculations.

Question 6 (10 marks)

Use a separate writing booklet

- (a) Is the function $f(x) = 8x^3 - 7x$ even, odd or neither? 2

Show working to support your answer.

$$\begin{aligned} f(-x) &= 8(-x)^3 - 7(-x) \\ &= -8x^3 + 7x & 1 \\ &= -(8x^3 - 7x) \\ &= -f(x) \end{aligned}$$

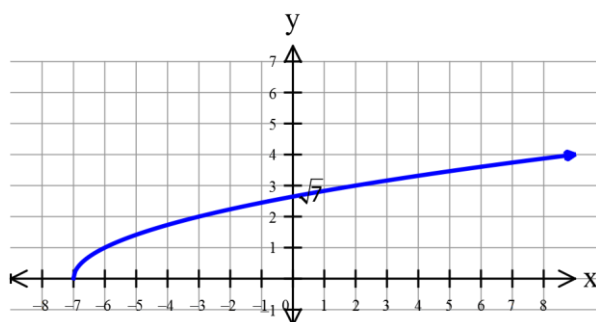
ODD function 1

- (b) Find the centre and radius of the circle with equation $x(x-2) + y(y+10) = 23$. 2

$$\begin{aligned} x^2 - 2x + y^2 + 10y &= 23 \\ x^2 - 2x + 1 + y^2 + 10y + 25 &= 23 + 1 + 25 \\ (x-1)^2 + (y+5)^2 &= 49 \\ \text{Centre } (1, -7) & & 1 \\ \text{Radius } r = 7 & & 1 \end{aligned}$$

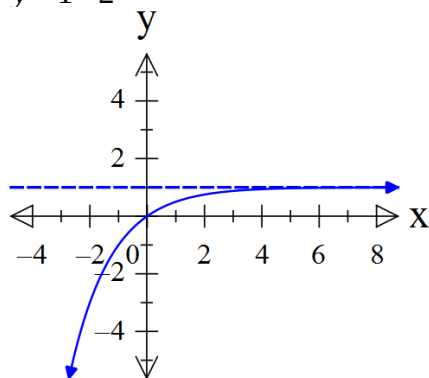
- (c) On separate diagrams, sketch each of the following, showing all essential features.

- (i) $y = \sqrt{x+7}$ 2



(ii) $y = 1 - 2^{-x}$

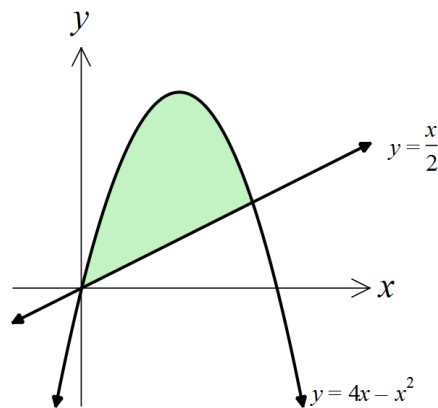
2



Students were penalised here (0.5 mk) if they didn't label a point (other than the origin)

Also it was necessary to draw the asymptote.

(d) The diagram shows the graphs of the functions $y = 4x - x^2$ and $y = \frac{1}{2}x$.



Write down a pair of inequalities that specify the shaded region.

2

$$y \leq 4x - x^2$$

1

$$y \geq \frac{1}{2}x$$

1

Students were penalised here (0.5 mk) if they used $<$ or $>$.

End of Question 6

Question 7 (11 marks)

Use a separate writing booklet

- (a) The function
- $y = f(x)$
- is defined as follows:

$$f(x) = \begin{cases} x-1 & \text{for } x \leq -2 \\ -1 & \text{for } -2 < x \leq 1 \\ x+1 & \text{for } x > 1 \end{cases}$$

Evaluate $f(-2) - f(17) + f(1)$.**1**

$$\begin{aligned} f(-2) - f(17) + f(1) &= (-2-1) - (17+1) + (-1) \\ &= -3-18-1 \\ &= -22 \end{aligned}$$

1

- (b) Solve each equation for
- $0^\circ \leq \theta \leq 360^\circ$
- , correct to the nearest degree where necessary.

(i) $\cos^2 \theta = \frac{3}{4}$

2

$$\cos^2 \theta = \frac{3}{4}$$

$$\cos \theta = \pm \frac{\sqrt{3}}{2}$$

1

$$\theta = 30^\circ$$

$$\theta = 30^\circ, 150^\circ, 210^\circ, 330^\circ.$$

1

(ii) $\tan 2\theta = -5$

2

$$\tan 2\theta = -5$$

$$\text{ref } \angle = 78^\circ 41'$$

1

tan negative in 2nd and 4th quadrants

$$2\theta = (180 - 78^\circ 41') \quad \text{or} \quad 2\theta = (360 - 78^\circ 41')$$

$$\text{or } 2\theta = 360 + 101^\circ 19' \quad \text{or } 2\theta = 360 + 281^\circ 19'$$

$$2\theta = 101^\circ 19'$$

$$\text{or } 2\theta = 281^\circ 19'$$

$$\text{or } 2\theta = 461^\circ 19'$$

$$\text{or } 2\theta = 641^\circ 19'$$

$$\theta = 50^\circ 40'$$

$$\text{or } \theta = 140^\circ 40'$$

$$\text{or } \theta = 230^\circ 40'$$

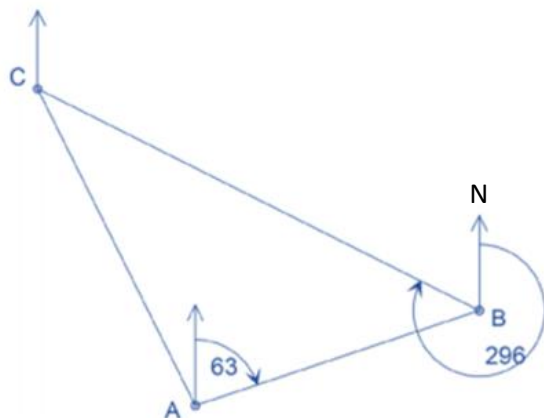
$$\text{or } \theta = 320^\circ 40'$$

1

- (c) A ship sails 50km from port A to port B on a bearing of 063° T and then sails 130km from port B to port C on a bearing of 296° T .

(i) Draw a diagram to represent this information.

1



(ii) Explain why $\angle ABC = 53^\circ$

$$\angle CBN + 296 = 360 \quad (\text{angles at a point})$$

$$\angle CBN = 64^\circ$$

$$\angle ABC + \angle CBN + 63^\circ = 180^\circ \quad (\text{cointerior angles on parallel lines})$$

$$\therefore \angle ABC = 180^\circ - 64 - 63$$

$$\angle ABC = 53^\circ$$

(or other solution - eg alternate angles)

(iii) Find, correct to the nearest km, the distance from port A to port C.

2

$$b^2 = 130^2 + 50^2 - 2 \times 130 \times 50 \times \cos 53^\circ$$

1

$$b^2 = 11576.4 \dots$$

$$b = 107.59$$

$$b = 108 \text{ km}$$

1

(iv) Find the bearing of port A from port C. Answer to the nearest degree.

2

$$\frac{\sin C}{50} = \frac{\sin 53}{107.59}$$

$$C = 22^\circ$$

1

$$\text{bearing} = 116 + 22$$

$$= 138^\circ$$

1

Students who used the SINE rule are reminded to check the ambiguous case, which worked in this question.

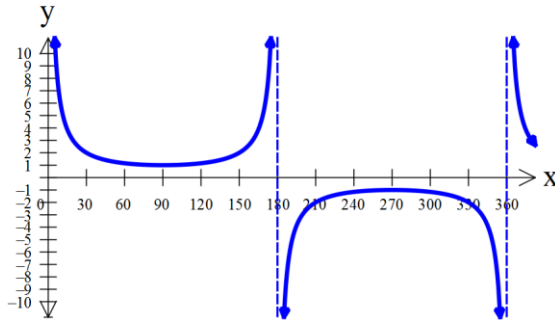
End of Question 7

Question 8 (10 marks)

Use a separate writing booklet

- (a) Draw a neat sketch of $y = \operatorname{cosec} x$ for $0^\circ \leq x \leq 360^\circ$.

2

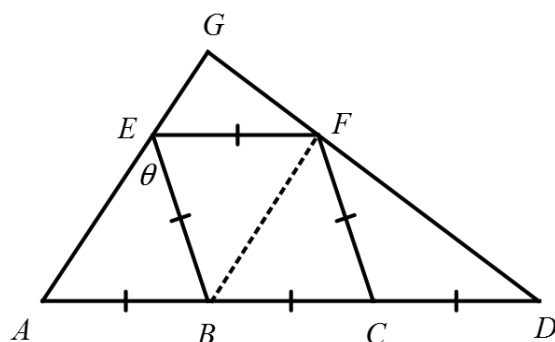


- (b) Prove that $\frac{\sin \beta}{1 + \cos \beta} = \frac{1 - \cos \beta}{\sin \beta}$.

2

$$\begin{aligned} LHS &= \frac{\sin \beta}{1 + \cos \beta} \\ &= \frac{\sin \beta}{1 + \cos \beta} \times \frac{1 - \cos \beta}{1 - \cos \beta} \\ &= \frac{\sin \beta(1 - \cos \beta)}{1 - \cos^2 \beta} \\ &= \frac{\sin \beta(1 - \cos \beta)}{\sin^2 \beta} \\ &= \frac{1 - \cos \beta}{\sin \beta} \\ &= RHS \end{aligned}$$

- (c) $ABCD$ is a straight line such that $AB = BC = CD$. $EFCB$ is a rhombus and $\angle AEB = \theta$.



NOT TO SCALE

- (i) Show that $\angle EBC = 2\theta$.

2

$$\angle EAB = \theta \quad (\text{equal angles opposite equal sides, } EB = AB)$$

1

$$\angle EBC = \angle BAE + \angle AEB \quad (\text{exterior angle of a triangle})$$

1

$$= \theta + \theta$$

$$= 2\theta$$

- (ii) The diagonal BF is drawn in the rhombus $EFCB$.

2

Show that $BF \parallel AE$.

$$\angle EBF = \angle FBC \quad (\text{diagonals of a rhombus bisect the vertex angles})$$

$$\angle EBF = \angle FBC = \theta$$

$$\angle AEB = \angle EBF \quad (\text{both } \theta)$$

AE is parallel to BF (alternate angles equal on parallel lines)

- (iii) Hence or otherwise, find the size of $\angle AGD$. Show all working

2

$$\angle BFC = \angle FBC \quad (\text{equal angles opposite equal sides})$$

$$\angle FCD = \angle BFC + \angle FBC \quad (\text{exterior angle of a triangle})$$

$$\angle FCD = 2\theta$$

$$\text{OR } \angle FCD = \angle EBC \quad (\text{corresponding angles, } FC \parallel EB)$$

$$\angle FCD = 2\theta$$

$$\angle FDC = \angle CFD \quad (\text{equal angles opposite equal sides})$$

$$\angle FCD + \angle FDC + \angle CFD = 180 \quad (\text{angle sum of triangle})$$

$$\angle FDC = \frac{180 - 2\theta}{2}$$

$$= 90 - \theta$$

$$\angle AGD + \theta + (90 - \theta) = 180 \quad (\text{angle sum of triangle})$$

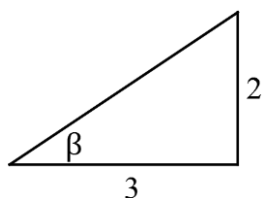
$$\angle AGD = 90^\circ$$

End of Question 8

Question 9 (10 marks)

Use a separate writing booklet

- (a) If β is a reflex angle such that $\tan \beta = \frac{2}{3}$, find the exact value of $\cos \beta$. **1**



tan is +ive so B is in 3rd quadrant (cos -ive)

$$\cos \beta = -\frac{3}{\sqrt{13}}$$

- (b) Solve $6\sin^2 x + 7\cos x - 1 = 0$ for $0^\circ \leq x \leq 360^\circ$. **3**
Give answers to the nearest degree.

$$6(1 - \cos^2 \theta) + 7\cos \theta - 1 = 0$$

$$-6\cos^2 \theta + 7\cos \theta + 5 = 0$$

$$6\cos^2 \theta - 7\cos \theta - 5 = 0$$

$$(3\cos \theta - 5)(2\cos \theta + 1) = 0$$

$$\cos \theta = \frac{5}{3} \quad \text{or} \quad \cos \theta = -\frac{1}{2}$$

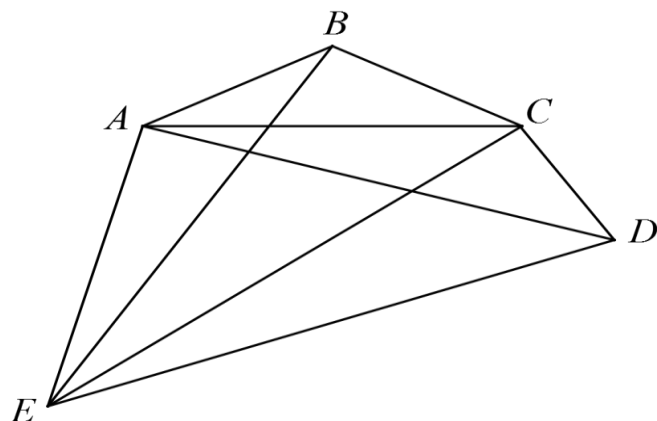
no solution

cos negative in 2nd / 3rd quadrants

$$\theta = 180 - 60^\circ, \quad 180 + 60^\circ$$

$$\therefore \theta = 120^\circ, 240^\circ$$

- (c) In pentagon $ABCDE$, $AB = BC = CD$, $\angle AEC = \angle ACE$ and $\angle BAE + \angle BAC = \angle BCD$.



NOT TO SCALE

- (i) Explain why $\angle BAC = \angle BCA$ 1

$$\angle BAC = \angle BCA \text{ (equal angles opposite equal sides, } BA = BC)$$

- (ii) Show that $\angle BAE = \angle ACD$. 2

$$\angle BAE + \angle BAC = \angle BCD \text{ given}$$

$$\angle BAE = \angle BCD - \angle BAC$$

$$\angle BAE = \angle BCD - \angle BCA \text{ (from i)}$$

$$\angle BAE = \angle ACD \text{ (} \angle BCA \text{ \& } \angle ACD \text{ are adjacent angles)}$$

- (iii) Hence, prove that $\triangle EAB \equiv \triangle ACD$ 3

In $\triangle EAB$ & $\triangle ACD$

$$AB = CD \text{ (given)}$$

$$\angle BAE = \angle ACD \text{ (from above (ii))}$$

$$AE = CA \text{ (equal sides opposite equal angles, given } \angle ACE = \angle AEC)$$

$$\therefore \triangle EAB \equiv \triangle ACD \text{ (SAS)}$$

End of Examination